CS341 · COMPUTER GRAPHICS

# Midterm 472 الحل

20 درجة

**محارب**

## 01 حل Q1: إجابات قصيرة

### 1. Why use aspect ratio?

Aspect ratios are useful for maintaining geometric features of figures drawn on a screen: a square remains a square and does not become a rectangle, and a circle stays a circle and does not become an ellipse.

### 2. Absolute vs relative coordinates

**Absolute**: the values are the actual positions within the coordinate system in use (measured from the origin).
**Relative**: the position is given as an offset from the last referenced position (the current position).
Example: current position (3, 8) + relative (2, −1) → absolute (5, 7).

### 3. Define a vector

A vector is a quantity that has **magnitude and direction** (like velocity). It represents a displacement and has no fixed location.

### 4. Vector from P to Q

```
v = Q − P
```

### 5. Rays emitted by a cathode ray tube

The rays are **electrons** (an electron beam). When they hit the phosphor coating of the screen, the phosphor emits **light**.

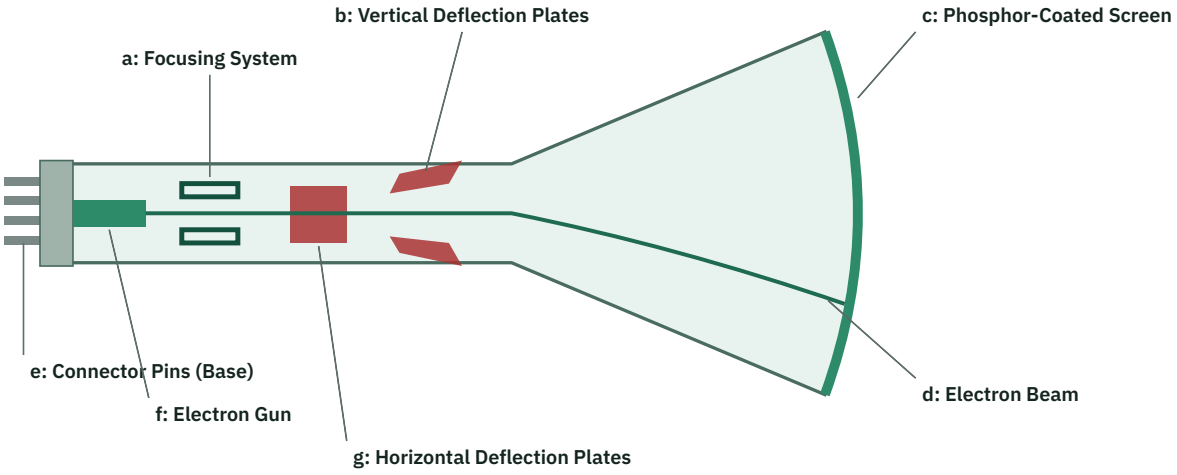
## 02 حل Q2: صح وخطأ

| # | الجواب | السبب |
|---|---|---|
| 1 | **True** | هذا تعريف المتجه. |
| 2 | **False** | مجموع نقطتين آخر إحداثي فيه 2، والنقطة الصحيحة تحتاج مجموع المعاملات = 1 (affine combination). |
| 3 | **True** | الزاوية 90° و `cos 90° = 0`، فالضرب النقطي = 0. |
| 4 | **False** | طول متجه الوحدة = 1 بالضبط. |

## 03 حل Q3: اختيار من متعدد

| # | الجواب | السبب |
|---|---|---|
| 1 | **B) Frame buffer** | في الـ raster الصورة مخزنة في الـ refresh (frame) buffer. |
| 2 | **B) Random scan** | في الـ random scan الشعاع يرسم الصورة مباشرة. الـ raster يمسح الشاشة سطر سطر. |
| 3 | **C) Dot pitch** | الوضوح يعتمد على resolution و dot pitch و refresh rate. |
| 4 | **C) Mouse** | الباقي أجهزة إخراج. |
| 5 | **D) Data encryption** | المهام الثلاث: Modeling و Rendering و Animation. |
| 6 | **D) All of the above** | الثلاثة كلها من تطبيقات السلايد. |
| 7 | **A** | creation, storage, and manipulation of models and images. |
| 8 | **A** | نفس التعريف. |

## 04 حل Q4: أجزاء الـ CRT




| الحرف | الجزء |
|---|---|
| a | Focusing System |
| b | Vertical Deflection Plates |
| c | Phosphor-Coated Screen |
| d | Electron Beam |
| e | Connector Pins (Base) |
| f | Electron Gun |
| g | Horizontal Deflection Plates |

## 05 حل Q5: DDA

$dx = 4 - 2 = 2 \qquad dy = 10 - 2 = 8$

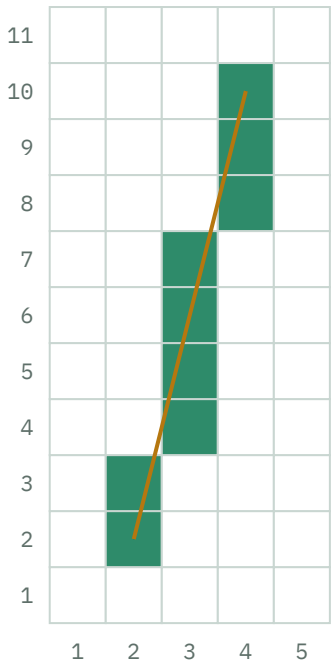
$m = \frac{8}{2} = 4 \rightarrow |m| > 1 \rightarrow \text{step in y}$

$steps = \max(|dx|, |dy|) = 8$

$xInc = \frac{2}{8} = 0.25 \qquad yInc = \frac{8}{8} = 1$

$round(a) = int(a + 0.5)$

| k | x | y | round(x) | round(y) |
|---|---|---|---|---|
| 0 | 2 | 2 | 2 | 2 |
| 1 | 2.25 | 3 | 2 | 3 |
| 2 | 2.5 | 4 | 3 | 4 |
| 3 | 2.75 | 5 | 3 | 5 |
| 4 | 3 | 6 | 3 | 6 |
| 5 | 3.25 | 7 | 3 | 7 |
| 6 | 3.5 | 8 | 4 | 8 |
| 7 | 3.75 | 9 | 4 | 9 |
| 8 | 4 | 10 | 4 | 10 |




Points: (2,2) (2,3) (3,4) (3,5) (3,6) (3,7) (4,8) (4,9) (4,10)